

Shen Zhen MTC Co., LTD

TEST REPORT

SCOPE OF WORK

RF Testing—MUDV50B0-90106-T2, MUDV50B0-90106-S2, MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB, 4T-C50HJ4020KB, **-*50*****(* can from 0 to 9, A to Z or blank), PR50U3421SKB.

REPORT NUMBER

260508031SZN-002

ISSUE DATE

[REVISED DATE]

21 May 2026 [-----]

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DTV_RFa

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RADIO PERFORMANCE VERIFICATION SUMMARY

Report No.: 260508031SZN-002

☒ DVB-T ☒ DVB-T2 ☐ DVB-S ☐ DVB-S2

Model: MUDV50B0-90106-T2, MUDV50B0-90106-S2, MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB, 4T-C50HJ4020KB, *-50*****(* can from 0 to 9, A to Z or blank), PR50U3421SKB. Product Description: TV/LCD LED TV/LED LCD TV Brand names: AMTC, Sharp, Polaroid	Applicant: Shen Zhen MTC Co., LTD MTC Industry Park, 1st Lilang Road, Xialilang community, Nanwan Street, Longgang district, Shenzhen, China Sample Receipt Date: 01 December 2022 Test Conducted Date: 01 December 2022 to 06 January 2023
☒ 1st TEST ☐ 2nd TEST	ALL TESTS WERE CONDUCTED IN ACCORDANCE WITH: * ETSI EN 303 340 V1.2.1 (2020-09)
Test Site and Location:	Intertek Testing Services Shenzhen Ltd. No. 101&201, Building B, No. 308 Wuhe Avenue, Zhangkengjing Community, GuanHu Subdistrict, LongHua District, ShenZhen, P.R. China.
Test Result	Pass Fail See Remark
* ETSI EN 303 340 V1.2.1 (2020-09)	☒ ☐ ☐
When determining the test conclusion, the following Measurement Uncertainty of test has been considered at a level of confidence of 95%.	

Prepared and Checked By:

Approved By:

Signature

Bruce Zheng
Project Engineer
21 May 2026 Date

Signature

Rode Liu
Project Engineer
21 May 2026 Date

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Intertek Testing Services Shenzhen Ltd.

No. 101&201, Building B, No. 308 Wuhe Avenue, Zhangkengjing Community, GuanHu Subdistrict, LongHua District, ShenZhen, P.R. China.
Tel: (86 755) 8601 6288 Fax: (86 755) 8601 6751

RADIO PERFORMANCE RESULTS CONCLUSION
(with Justification)

RE: Radio Performance Testing Pursuant to Radio Equipment Directive (2014/53/EU)
Article 3.1(a) health, 3.2 Performed on the TV/LCD LED TV/LED LCD TV,

Model: MUDV50B0-90106-T2

MUDV50B0-90106-S2, MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB, 4T-C50HJ4020KB, **-*50*****(* can from 0 to 9, A to Z or blank), PR50U3421SKB.

We tested the TV/LCD LED TV/LED LCD TV, Model: MUDV50B0-90106-T2, to determine if it was in compliance with the relevant EN standards as marked on the Radio Performance Verification Summary. We found that the unit met the requirement of EN 62479, ETSI EN 303 340 standards when tested as received.

The Model: MUDV50B0-90106-S2, MUDV50B0-90106, MUDV50****-90106-## (*, # can from 0 to 9, A to Z or blank), 4T-C50FJ2KL1FB, 4T-C50FJ2EL2NB, 4T-C50HJ4020EB, 4T-C50HJ4020KB, **-*50*****(* can from 0 to 9, A to Z or blank), PR50U3421SKB are the same as the Model: MUDV50B0-90106-T2 in hardware aspect. The difference in model number serves as marketing strategy.

This report is based on previous report with report number 221201030SZN-002 dated 01 April 2023 (original signatory: Bruce Zheng, Rode Liu on file). Owing to adding the panel model HF500QUB, also removing the DVB-S/S2 interface, no test was required after evaluation.

The production units are required to conform to the initial sample as received when the units are placed on the market.

LABORATORY MEASUREMENTS**Configuration Information**

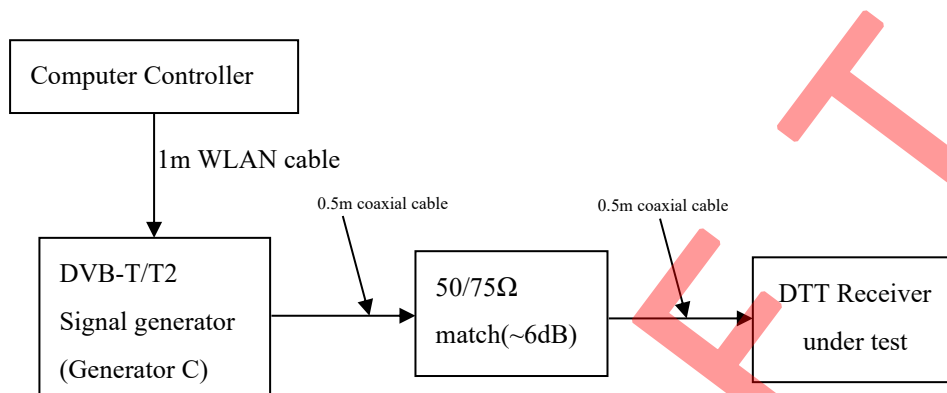
Equipment Under Test (EUT):	TV/LCD LED TV/LED LCD TV
Model:	MUDV50B0-90106-T2
Serial No.:	N/A
Support Equipment:	DTV Multisignal Generator(EIDEN MSD5000A)
Supported Broadcast Frequency Bands:	VHF & UHF
If Broadcast Receiver With Optional Preampfier:	No
Rated Voltage:	220-240V~, 50/60Hz, 128W

ETSI EN 303 340 ANNEX E APPLICABLE TESTS FOR DIFFERENT RECEIVER VARIANTS

Applicable tests				
Wanted signal configuration supported by receiver	Sensitivity tests clause 4.2.3	Adjacent channel selectivity tests clause 4.2.4 (see note 1 and note 3)	Blocking tests clause 4.2.5 (see note 4)	Overloading tests clause 4.2.6 (see note 5)
DVB-T only, VHF & UHF	Test-1 (DVB-T) Test-2 (DVB-T)	Test-1 (DVB-T) & (DVB-T2) Test-2 (DVB-T) & (DVB-T2) Test-3 (DVB-T) & (DVB-T2) Test-4 (DVB-T) Test-5 (DVB-T)	Test-1 (DVB-T)	Test-1 (DVB-T) (for broadcast receivers with optional preamplifier) Or Test-3 DVB-T (for broadcast receivers without optional preamplifier)
DVB-T & T2, VHF & UHF	Test-1 (DVB-T) Test-2 (DVB-T) Test-1 (DVB-T2) Test-2 (DVB-T2)			
DVB-T only, UHF only	Test-2 (DVB-T)			
DVB-T & T2, UHF only	Test-2 (DVB-T) Test-2 (DVB-T2)			
DVB-T2 only, VHF & UHF	Test-1 (DVB-T2) Test-2 (DVB-T2)	Test-1 (DVB-T2) Test-2 (DVB-T2) Test-3 (DVB-T2) Test-4 (DVB-T2) Test-5 (DVB-T2)	Test-2 (DVB-T2)	Test-1 (DVB-T2) (for broadcast receivers with optional preamplifier) Or Test-3 (DVB-T2) (for broadcast receivers without optional preamplifier)
DVB-T2 only, UHF only	Test-2 (DVB-T2)			
NOTE 1: For broadcast receivers that do not receive DVB-T/T2 signals above 698 MHz, Adjacent channel selectivity test 1 is not applicable. NOTE 2: Void. NOTE 3: Test-4 (DVB-T) and Test-5 (DVB-T) are UHF tests which apply to broadcast receivers capable of either DVB-T only, or DVB-T&T2 reception in UHF/VHF because DVB-T and DVB-T2 adjacent channel selectivities are correlated and also UHF/VHF selectivities are correlated. NOTE 4: Test-1 (DVB-T) is a UHF test which applies to broadcast receivers capable of either DVB-T only, or DVB-T&T2 reception in UHF/VHF because DVB-T and DVB-T2 blocking levels are correlated and UHF/VHF blocking levels are correlated. Test-2 (DVB-T2) is a UHF DVB-T2 test which applies to broadcast receivers only capable of DVB-T2 reception in UHF/VHF because UHF/VHF blocking levels are correlated. NOTE 5: Test-1 (DVB-T) and Test-3 (DVB-T) are UHF tests which apply to broadcast receivers capable of either DVB-T only, or DVB-T&T2 reception in UHF/VHF because DVB-T and DVB-T2 overload levels are correlated and UHF/VHF overload levels are correlated. Test-1 (DVB-T2) and Test-3 (DVB-T2) are UHF tests which apply to broadcast receivers only capable of DVB-T2 reception in UHF/VHF because UHF/VHF overload levels are correlated.				

ETSI EN 303 340 RECEIVER SENSITIVITY

Test Arrangement Description



Test Procedure

- a) The wanted signal, provided by signal generator-C, shall be set to the wanted signal frequency as shown in table 4 and configured with the appropriate DTT mode parameters.
- b) Determine the attenuation of the cables, splitters, 50/75 ohm matching pad.
- c) Set the level of the wanted signal C at the broadcast receiver tuner port to -50 dBm and note the value of the "attenuator C" (= AC1).
- d) Tune the DTT receiver to the wanted channel.
- e) Increase the "attenuator C" sufficiently to cause complete picture failure.
- f) Adjust "attenuator C" gradually until the onset of picture degradation occurs. Force the receiver under test to re-acquire and re-adjust "attenuator C" until the onset of picture degradation occurs. Note the attenuator setting (= AC2).
- g) The receiver sensitivity shall be recorded in the measurement record as:

$$-50 - (AC2 - AC1) \text{ dBm}$$
- h) Repeat step a) to g) for the remaining frequencies in EN 303 340 table 4.

Limits

Test number	Test description	C wanted signal centre frequency (MHz)	Required sensitivity limit (dBm)	
			DVB-T	DVB-T2
1	Sensitivity VHF	198,5	-77	-75
2	Sensitivity UHF	666	-77	-75

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ070-15	Matching Pad	R & S	RAM	2022-05-16	2023-05-16
SZ070-16	Combiner	Mini-Circuits	ZFSC-2-2500-S+	2022-10-17	2023-10-17
SZ180-12	DTV Multi Signal Generator	Eiden	MSD5000A	2022-04-01	2023-04-01

Note: Uncertainty: $\pm 2.8\text{dB}$ at a level of confidence of 95%.

Test Data

Test Temperature: 25°C Test Humidity: 50%

DVB-T

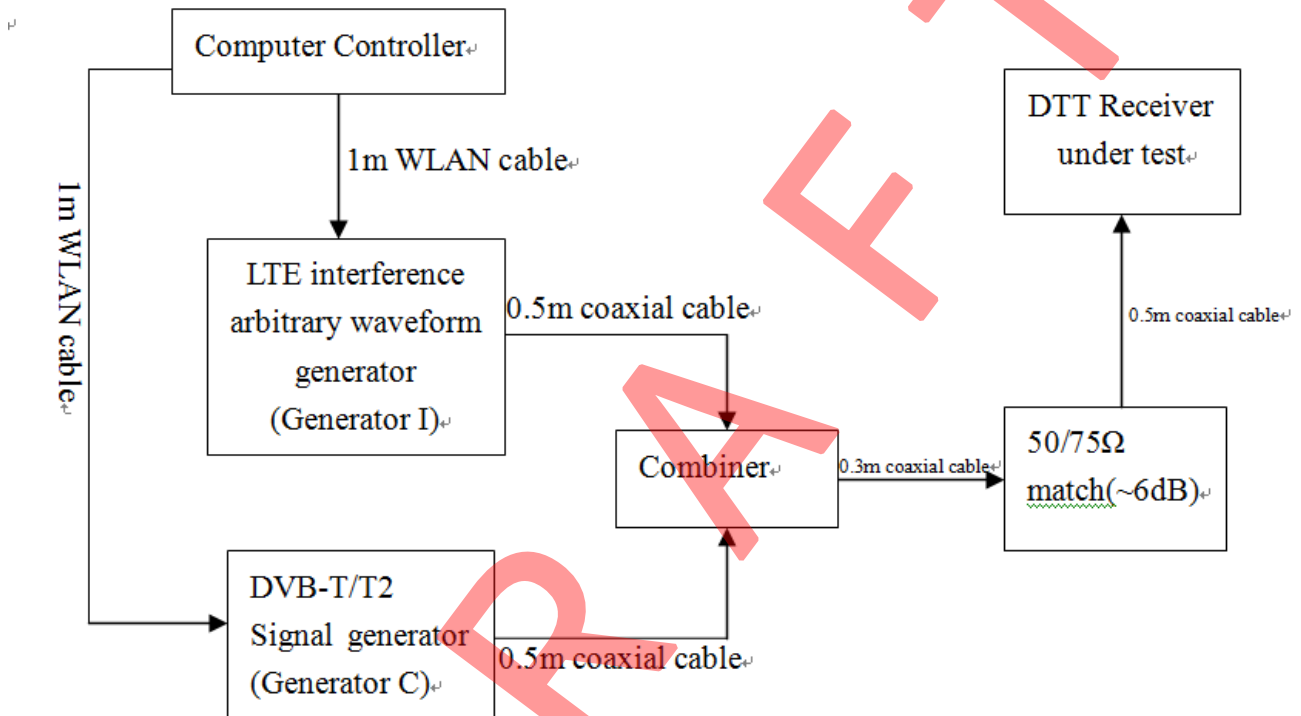
Test	C frequency [MHz]	Crms [dBm]	Crms Limit [dBm]	Margin [dB]	Verdict
1	198.5	-82.0	-77.0	5.0	PASS
2	666.0	-83.0	-77.0	6.0	PASS

DVB-T2

Test	C frequency [MHz]	Crms [dBm]	Crms Limit [dBm]	Margin [dB]	Verdict
1	198.5	-81.0	-75.0	6.0	PASS
2	666.0	-81.0	-75.0	6.0	PASS

ETSI EN 303 340 ADJACENT CHANNEL SELECTIVITY

Test arrangement description



Limits

Test	Interferer (I) type	I Interference test signal waveform name	C wanted signal centre frequency (MHz)	I interferer centre frequency (MHz)	I _{lic} (licensed) (dBm)	I _{rms} (rms) (dBm)	I LAPR (dB)	Minimum required I/C limit (where I = I _{lic}) (dB)		Equivalent wanted level Crms (dBm)	
								DVB-T	DVB-T2	DVB-T	DVB-T2
1 (see notes 1 and 2)	10 MHz LTE 800 BS light load (near idle)	LTE_BS-idle_V3_synth	786	796	-15	-23.3	8.3	35	36	-50	-51
2 (see notes 2)	10 MHz LTE 700 BS light load (near idle)	LTE_BS-idle_V3_synth	690	763 (see note 3)	-15	-23.3	8.3	43	43	-58	-58
3 (see notes 2)	10 MHz LTE 700 UE Video-Stream	Short_UE-Video-Stream_V2	690	708	-25	-42.7	17.7	33	38	-58	-63
4 (see notes 2)	N-1 UHF	8 MHz DVB-T	482	474	-30	-30	0	25	25	-55	-55
5 (see notes 2)	N+1 UHF	8 MHz DVB-T	482	490	-30	-30	0	25	25	-55	-55

NOTE 1: For broadcast receivers that do not receive DVB-T/T2 signals above 698 MHz, test 1 is not applicable.

NOTE 2: For applicability of tests see annex E.

NOTE 3: It is acceptable to use an alternative lower interference frequency such as 761 MHz. This may be necessary due to test equipment limitations.

Test Procedure

- a) Two signal generators, I and C, shall be connected to the receiver via a combining network.
- b) Determine the attenuation of the cables, combiner, splitter and 50/75 ohm matching pad.
- c) The wanted signal, provided by signal generator C, shall be set to the wanted signal frequency C as shown in EN 303 340 table 5 and configured with the appropriate DTT mode parameters. Turn this generator off (or to at least 30 dBc below the interferer signal) maintaining output impedance.
- d) The unwanted signal I, provided by signal generator I, shall be configured with the required interferer waveform and interferer signal frequency as shown in EN 303 340 table 5.
- e) The unwanted signal power at the broadcast receiver tuner port shall be set to the rms interferer power I_{rms} in EN 303 340 table 5. This power shall be verified with the power meter or spectrum analyser. Recommended instrument settings are described in EN 303 340 clause D.2.
- f) Turn on signal generator C and adjust "attenuator C" so that the rms powers of the interferer I and wanted signal C are equal. Note this attenuator setting ($= AC1$).
- g) Tune the DTT receiver to the wanted channel.
- h) Increase "attenuator C" in 1 dB steps or less until the onset of picture degradation occurs. Force the receiver under test to re-acquire and re-adjust "attenuator C" until the onset of picture degradation occurs. Note the attenuator setting ($= AC2$).
- i) The adjacent channel selectivity (I/C) shall be recorded in the measurement record as $AC2 - AC1 + LAPR$ dB, where LAPR is the licensed to average power ratio shown in EN 303 340 table 5.
Alternatively the measured wanted signal power C shall be recorded in the measurement record as $I_{rms} - (AC2 - AC1)$ dBm.
- j) Repeat step b) to i) for the remaining tests in EN 303 340 table 5.

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ065-08	Wideband Radio Communication Tester	R & S	CMW500	2022-07-04	2023-07-04
SZ070-15	Matching Pad	R & S	RAM	2022-05-16	2023-05-16
SZ070-16	Combiner	Mini-Circuits	ZFSC-2-2500-S+	2022-10-17	2023-10-17
SZ180-12	DTV Multi Signal Generator	Eiden	MSD5000A	2022-04-01	2023-04-01
SZ180-14	Vector Signal Generator	R & S	SMU200A	2022-05-16	2023-05-16

Note: Uncertainty: $\pm 2.8\text{dB}$ at a level of confidence of 95%.

Test Data

Test Temperature: 25°C Test Humidity: 50%

DVB-T

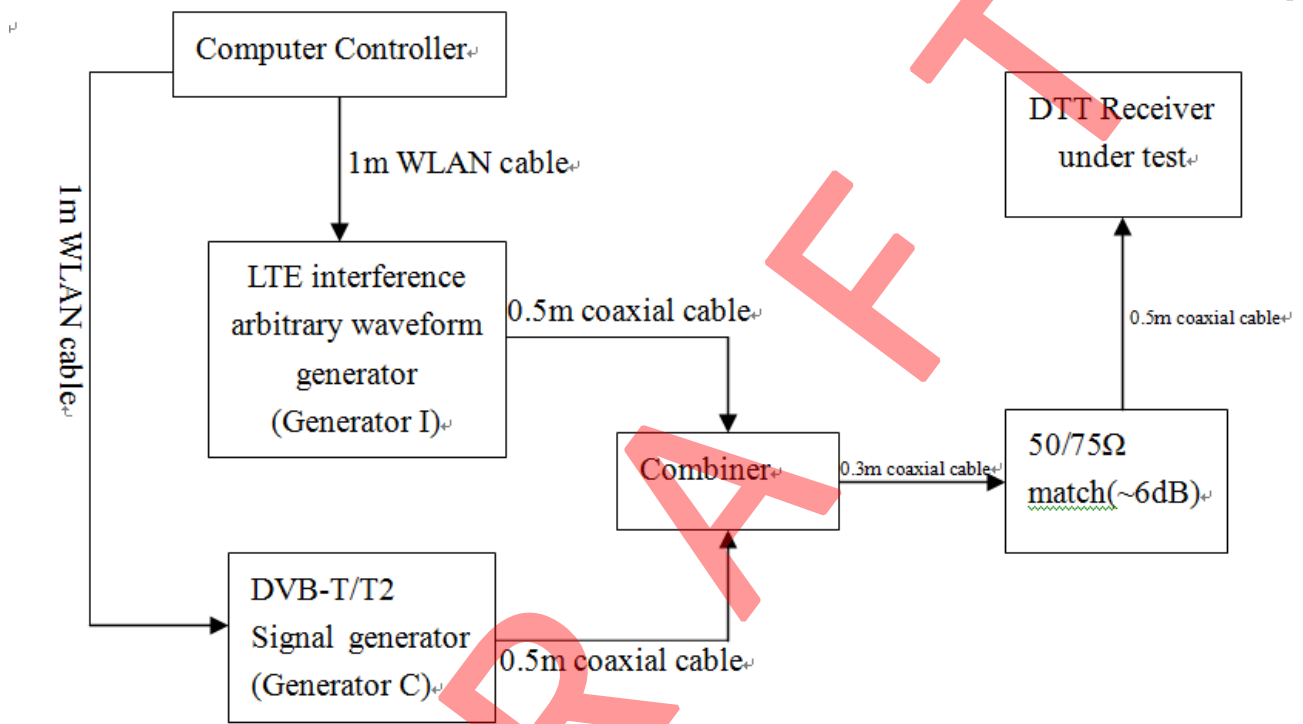
Test	C frequency [MHz]	I frequency [MHz]	Irms [dBm]	Crms [dBm]	Crms Limit [dBm]	Margin [dB]	Verdict
1	786.0	796.0	-23.3	-56.3	-50	6.3	PASS
2	690.0	763.0	-23.3	-64.3	-58	6.3	PASS
3	690.0	708.0	-42.7	-64.7	-58	6.7	PASS
4	482.0	474.0	-30.0	-62	-55	7.0	PASS
5	482.0	490.0	-30.0	-61	-55	6.0	PASS

DVB-T2

Test	C frequency [MHz]	I frequency [MHz]	Irms [dBm]	Crms [dBm]	Crms Limit [dBm]	Margin [dB]	Verdict
1	786.0	796.0	-23.3	-57.3	-51.0	6.3	PASS
2	690.0	763.0	-23.3	-64.3	-58.0	6.3	PASS
3	690.0	708.0	-42.7	-69.7	-63.0	6.7	PASS

ETSI EN 303 340 BLOCKING

Test arrangement description



Limits

Test	Interferer (I) type	I interference test signal waveform name	C wanted signal centre frequency (MHz)	I centre frequency (MHz)	C _{rms} (rms) (dBm)	Required blocking level I _{rms} (dBm)	
						DVB-T	DVB-T2
1	10 MHz LTE 700 BS fully loaded	LTE_BS-100PC_synth	690	763	-71	-25	
2	10 MHz LTE 700 BS fully loaded	LTE_BS-100PC_synth	690	763	-69		-25

Test Procedure

- a) Two signal generators, I and C, shall be connected to the receiver via a combining network.
- b) Determine the attenuation of the cables, combiner, splitter and 50/75 ohm matching pad.
- c) The wanted signal, provided by signal generator C, shall be set to the wanted signal frequency as shown in EN 303 340 table 6 and configured with the appropriate DTT mode parameters.
- d) The unwanted signal I, provided by signal generator I, shall be configured with the required interferer waveform, interferer signal frequency as shown in EN 303 340 table 6.
- e) Adjust "attenuator C" to set the wanted level to -40 dBm at the broadcast receiver tuner port. This power shall be verified with the power meter or spectrum analyser. Then reduce the wanted signal power to the value of Crms in EN 303 340 table 6 by increasing "attenuator C" by 31 dB for test 1 or by 29 dB for test 2.
- f) Adjust "attenuator I" to set the interferer level to -40 dBm at the broadcast receiver tuner port. This power shall be verified with the power meter or spectrum analyser. Then reduce the interferer signal power to that of the wanted signal (Crms in EN 303 340 table 6) by increasing "attenuator I" by 31 dB for test 1 or by 29 dB for test 2. Note the value of "attenuator I" (= AI1).
- g) Tune the DTT receiver to the wanted channel.
- h) Reduce "attenuator I" until the onset of picture degradation occurs. Force the receiver under test to re-acquire and re-adjust "attenuator I" until the onset of picture degradation occurs. Note the attenuator setting (= AI2).
- i) The blocking level shall be recorded in the measurement record as $Crms + AI1 - AI2$ dBm where Crms is the value given in EN 303 340 table 6.

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ065-08	Wideband Radio Communication Tester	R & S	CMW500	2022-07-04	2023-07-04
SZ070-15	Matching Pad	R & S	RAM	2022-05-16	2023-05-16
SZ070-16	Combiner	Mini-Circuits	ZFSC-2-2500-S+	2022-10-17	2023-10-17
SZ180-12	DTV Multi Signal Generator	Eiden	MSD5000A	2022-04-01	2023-04-01
SZ180-14	Vector Signal Generator	R & S	SMU200A	2022-05-16	2023-05-16

Note: Uncertainty: $\pm 2.8\text{dB}$ at a level of confidence of 95%.

Test Data

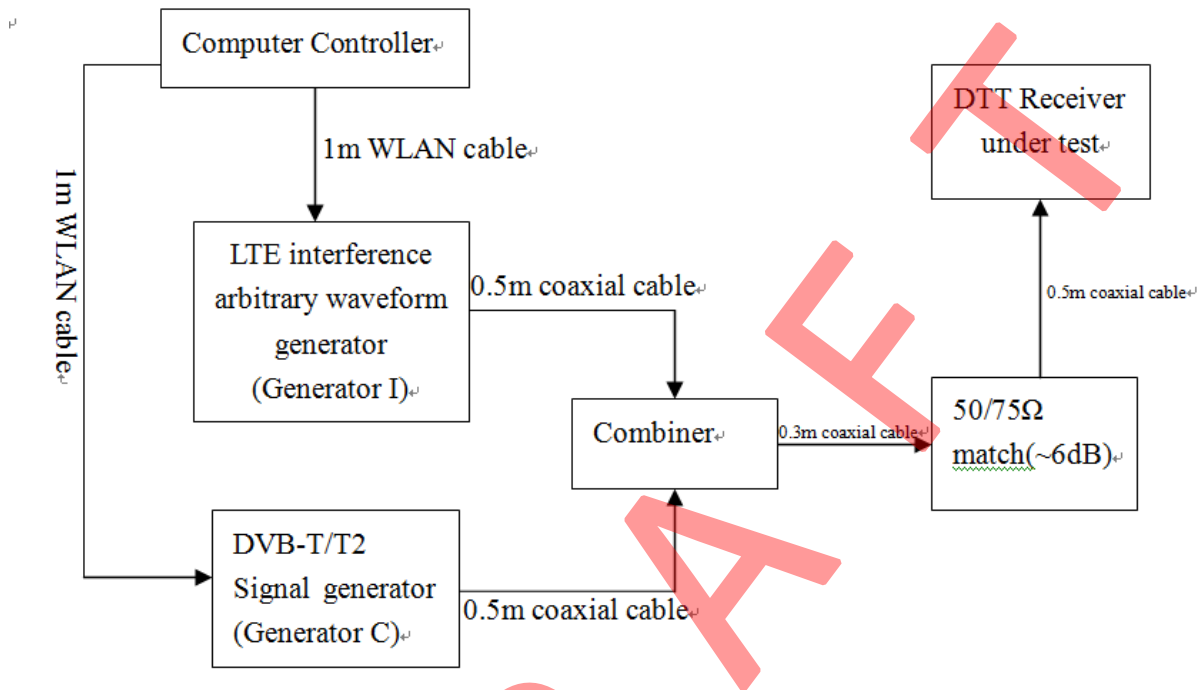
Test Temperature: 25°C Test Humidity: 50%

DVB-T

Test	C frequency [MHz]	I frequency [MHz]	Crms [dBm]	Irms [dBm]	Irms Limit [dBm]	Margin [dB]	Verdict
1	690.0	763.0	-71.0	-20.0	-25.0	5.0	PASS

ETSI EN 303 340 OVERLOADING

Test arrangement description



Limits

Test	Interferer (I) type	I interference test signal waveform name	C wanted signal centre frequency (MHz)	I centre frequency (MHz)	C _{rms} (rms) (dBm)	Required overload level I _{rms} for DTT configurations in tables 2 and 3 (dBm)	
						DVB-T	DVB-T2
1 Broadcast receivers with optional preamplifier (see notes 1, 2 and 4)	10 MHz LTE 700 BS fully loaded	LTE_BS-100PC_synth	690	763 (see note 1)	-35	-8	-8
2 Void							
3 Broadcast receivers without optional preamplifier (see notes 1, 2 and 4)	10 MHz LTE 700 BS fully loaded	LTE_BS-100PC_synth	690	763 (see note 1)	-35	-4	-4

NOTE 1: It is acceptable to test compliance at an alternative lower interference frequency (e.g. 761 MHz). This may be necessary due to test equipment limitations.

NOTE 2: For applicability of tests see annex E.

NOTE 3: Void.

NOTE 4: For the tests in the present table, "optional preamplifier" means an additional user controllable amplifier before the actual tuner IC device in the broadcast receiver, which may be available to the user and documented in the user documentation. The manufacturer shall declare the presence of an optional preamplifier in the declaration of conformity.

Test Procedure

- a) Two signal generators, I and C, shall be connected to the receiver via a combining network.
- b) Determine the attenuation of the cables, combiner, splitter and 50/75 ohm matching pad.
- c) The wanted signal, provided by signal generator C, shall be set to the wanted signal frequency as shown in EN 303 340 table 7 and configured with the appropriate DTT mode parameters
- d) The unwanted signal I, provided by signal generator I, shall be configured with the required interferer waveform, interferer signal frequency as shown in EN 303 340 table 7.
- e) Adjust "attenuator C" to set the wanted level to C_{rms} at the broadcast receiver tuner port.

This

power shall be verified with the power meter or spectrum analyser.

- f) Adjust "attenuator I" to set the interferer level to same level as C_{rms} at the broadcast receiver tuner port. This power shall be verified with the power meter or spectrum analyser. Note the value of "attenuator I" ($= A_{I1}$).
- g) Tune the DTT receiver to the wanted channel.
- h) Reduce "attenuator I" until the onset of picture degradation occurs . Force the receiver under test to re-acquire and re-adjust "attenuator I" until the onset of picture degradation occurs.

Note

the attenuator setting ($= A_{I2}$).

- i) The overload level shall be recorded in the measurement record as $C_{rms} + A_{I1} - A_{I2}$ dBm where C_{rms} is the value shown in EN 303 340 table 7.

Used Test Equipment

Equip No.	Description	Manufacturer	Model No.	Cal. Date	Due Date
SZ065-08	Wideband Radio Communication Tester	R & S	CMW500	2022-07-04	2023-07-04
SZ070-15	Matching Pad	R & S	RAM	2022-05-16	2023-05-16
SZ070-16	Combiner	Mini-Circuits	ZFSC-2-2500-S+	2022-10-17	2023-10-17
SZ180-12	DTV Multi Signal Generator	Eiden	MSD5000A	2022-04-01	2023-04-01
SZ180-14	Vector Signal Generator	R & S	SMU200A	2022-05-16	2023-05-16

Note: Uncertainty: $\pm 2.8\text{dB}$ at a level of confidence of 95%.

Test Data

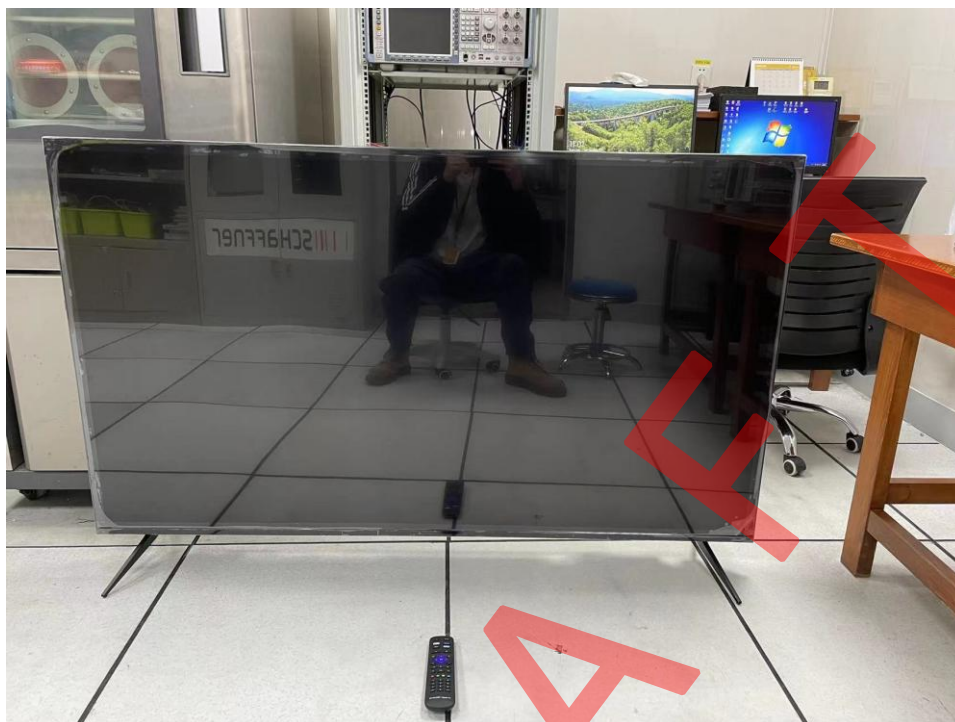
Test Temperature: 25°C Test Humidity: 50%

DVB-T

Test	C frequency [MHz]	I frequency [MHz]	Crms [dBm]	Irms [dBm]	Irms Limit [dBm]	Margin [dB]	Verdict
3	690.0	763.0	-35.0	0.0	-4.0	4.0	Pass

PHOTOS OF TEST SET-UP

Receiver Sensitivity & Adjacent Channel Selectivity & Blocking & Overloading



PHOTOS OF EUT

Please refer to Annex of EUT Photos_260508031SZN.

***** End of Report*****

DRAFT